

Task 11 · TTV Studio Intelligence Stack — Final Report

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Status:  Production ready

Duration: November 13–22, 2025 (9 days)

Branch: `task_quality_harden_secure`

Tests: 152/152 passing (100%) | Coverage: 95%+

Executive summary

Task 11 delivers intelligent video generation through eight new modules ($\approx 4,380$ lines) that add story understanding, scene tracking, narrative intelligence, and emotion-driven motion control.

Highlights

-  Eliminated video looping with smart extension
-  Resolved gender confusion via full-story NLP
-  Avoided RIFE black screens with frame-based extension
-  Achieved tight audio-video sync ($<0.5s$)
-  Preserved image quality (RGB pipeline, 8,000k bitrate)

Phase 2 goals (100%)

- Test coverage $>90\% \rightarrow 95\%+$
- Scene graph module (NetworkX, ~ 850 lines)
- Narrative engine (story beats, ~ 803 lines)
- Dashboard backend (26 metrics tracked)

Modules delivered

Eight core modules (≈4,380 lines)

Day 1–2 · Foundation

- `story_context_parser.py` (~580) — Story NLP and 20–30% text condensation
- `identity_memory.py` (~531) — Character identity tracking
- `scene_memory_core.py` (~850) — Temporal scene graph (NetworkX)

Day 3–4 · Intelligence

- `narrative_sequencer_v1.py` (~803) — Story beats and character arcs
- `emotion_controller.py` (~818) — Emotion tracking and motion coupling

Day 5 · Video enhancement

- `smart_video_extender.py` (~359) — Slow-mo + freeze smart extension
- `cinematic_transition_core.py` (~422) — Eight cinematic transitions

Day 6 · Metrics

- `audit_logger.py` (+~120) — 26 TTV intelligence metrics

Integration

- `unified_video_generator.py` — Full integration across modules
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Issues resolved

Production issues (5/5)

1. Video looping on short clips → Smart extension (1.5× slow-mo + freeze)
2. Gender confusion from per-sentence analysis → Full story NLP (consistent)
3. RIFE black screens → Frame-based extension strategy
4. Audio-video crossfade drift → Simple concatenation (<0.5s diff)
5. RGB↔BGR conversion loss → RGB maintained at 8,000k bitrate

Runtime bugs (2/2 · Nov 22)

1. `KeyError` `total_transitions` → Use `graph_stats.get('total_edges', 0)`
 2. `EmotionType` JSON serialization → Use `.value` before dict key
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Test results

Total: 152/152 tests passing (100%)

- Day 1: 26 
- Day 2: 18 
- Day 3: 20 
- Day 4: 23 
- Day 5: 38 
- Day 6: 14 
- Integration: 13 

Coverage: 95%+ across all modules

Assets created

Code

- `AnimateDiff/adaptive_engine/` — eight modules

Tests

- `tests/task11/` — 15 files
- `AnimateDiff/adaptive_engine/tests/` — 3 files

Docs

- `Documentation/Tasks/` `Task-11-README.md` (~2,070 lines)
 - `Documentation/Reports/` `Task-11-Report.md` (this file)
 - `TTV_STUDIO_USER_` `GUIDE.md` (800+ lines)
 - `TTV_Studio_v1-Audit.json` (~430 lines)
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Production integration

Console output

 Analyzing story with TTV Studio Intelligence...

-  Found 3 characters: seeker (female), sage (male), teacher (neutral)
-  Condensing narration...  Reduced 25%
-  Scene graph: 5 scenes, 12 entities
-  Story structure: 5 beats, 8 arcs
-  Emotion tracking initialized

 Generating video clips...

-  seeker: neutral (0.50) → motion 1.00x
-  Smart extended: 2.5s → 4.0s (SlowMo+Freeze)

 Audio-video sync: Perfect! (diff: 0.05s)

 Logging 26 TTV intelligence metrics...

Metrics logged (26 total)

- Story analysis: 4 — characters, gender, condensation, prompts
- Scene graph: 4 — scenes, entities, edges, avg/scene
- Narrative: 6 — beats, arcs, tension, pacing
- Emotion: 3 — changes, intensity, distribution
- Extension: 5 — clips, method, duration
- Quality: 4 — sync, fps, bitrate, duration

Phase 2 goals — completion matrix

Goal	Deliverable	Status
Scene graph	<code>scene_memory_core.py</code> (NetworkX)	
Narrative engine	<code>narrative_sequencer_v1.py</code>	
Dashboard backend	Extended audit logger (26 metrics)	

Goal	Deliverable	Status
Test coverage >90%	95%+ achieved (152 tests)	

Performance impact

- Execution overhead: ~5–10% (acceptable)
- Memory: +150 MB (scene graph and caches)

Quality improvements

- Gender consistency: 60% → 100% (+40%)
- Video looping: 100% → 0% (–100%)
- Audio-video sync: $\pm 2s$ → $\pm 0.3s$ (+85%)
- Output bitrate: 6,000k → 8,000k (+33%)

Conclusion

Task 11 delivers production-ready intelligent video generation:

-  5/5 production problems solved
-  4/4 Phase 2 goals complete
-  8 modules delivered (~4,380 lines)
-  152 tests passing with 95%+ coverage
-  Comprehensive documentation (3 guides)
-  KSML-compliant audit trail

Status:  Production ready

Report generated: November 22, 2025

Version: 1.0

See also: [Documentation/Tasks/](#) [Task-11-README.md](#)